

Introduction to Automata Theory

Problem Set 1

Daibik Barik, Shankha Suvra Dam

May 23, 2026

1. Design a DFA that accepts the string if the binary number it represents is divisible by 3.

2. Let

$$A = \{w \mid w \text{ starts with a } 0\}$$

and

$$B = \{w \mid w \text{ starts with a } 1\}.$$

Describe the language $A \circ B$ in plain english. Can you construct a DFA for it and show the similarities with the DFAs of A and B ?

3. Suppose a language L contains a finite number of strings (for example, exactly 1,000 distinct strings, each of varying lengths). Is it guaranteed to be a Regular Language? Can you always build a DFA for any finite set of words, or is there a catch?!
4. Design a DFA where a string is accepted if and only if:
 - the number of 0s is divisible by 2, and
 - the number of 1s is divisible by 3.
5. If I tell you that a mysterious language L satisfies the property

$$L = L \circ L,$$

what can you guess about the strings inside L ? Can you think of a non-empty example of such a language?

6. Suppose you have a DFA with k states over an alphabet of size

$$|\Sigma| = n.$$

If you convert it to a formal mathematical description, what is the exact number of transition rules you must define in the mapping of

$$\delta?$$

7. Let L be a regular language recognized by a DFA M . We define the reverse language L^R as the set of all strings in L read backward (e.g., if 011 is in L , then 110 is in L^R).

If you take a standard DFA diagram and simply flip the direction of every single transition arrow, and turn the start state into an accept state (and vice versa), why does the resulting diagram fail to be a valid DFA?

Even though the flipped diagram breaks the rules of being deterministic (since states might now have multiple arrows for the same character, or none at all), the language L^R is still guaranteed to be regular. Try cooking up a reasoning.